Sample Test 5
Times: 90 minutes

Question 1 (2 points). Let

$$\phi \equiv \forall x\Big(P(x, f(y)) \to \exists z\, Q(x, z, g(y, z))\Big)$$

and

$$\psi \equiv \exists x \forall y\Big(P(x, y) \to R(x, g(x, y))\Big) \wedge \neg Q(x, y).$$

a) Draw parse trees and identify the free and bound variables for ϕ and ψ.

b) Determine $\phi[h(y)/y]$ and $\psi[f(z)/x]$.

Question 2 (2 points). Verify the following sequents using natural deduction rules.

a) $\forall x \forall y\, P(x, y) \vdash \forall y \forall x\, P(x, y)$

b) $\forall x\big(P(x) \to Q(x)\big),\ \forall x\, P(x) \vdash \forall x\, Q(x)$

Question 3 (2 points). For each of the formulas below, find a model that satisfies the formula and one that does not.

a) $\exists x \forall y\, P(x, y) \wedge \forall y\, \neg P(a, y)$

b) $\forall x\big(P(x) \to \exists y\, Q(x, y)\big)$

Question 4 (2 points). Which of the following semantic entailments hold?

a) $\forall x\, P(x) \to \exists x\, Q(x) \models \exists x\big(P(x) \to Q(x)\big)$

b) $\forall x\big(P(x) \to Q(x)\big) \models \big(\forall x\, P(x)\big) \to \big(\forall x\, Q(x)\big)$

Question 5 (2 points). Use the tableau algorithm to determine whether the following formulas are valid or not. If it is not valid, is it satisfiable?

a) $\phi \equiv \big((p \vee q) \wedge (p \to r) \wedge (q \to r)\big) \to r$

b) $\psi \equiv \big(\forall x \exists y\, P(x, y)\big) \to \big(\exists y \forall x\, P(x, y)\big)$